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Education

🏛 University of Tehran

- M.Sc. in Computer Engineering (Computer Architecture)
- Cumulative GPA: **19.18/20**
- Thesis: *Designing Generative Models for Developing Deep Learning Applications in Video Compression*

Tehran, Iran
Sep. 2019 – Feb. 2023

🏛 University of Kashan

- B.Sc. in Computer Engineering (Software Engineering)
- GPA (last **two** years): **18.25/20** | • Cumulative GPA: **17.68/20**
- Thesis: *A Hybrid Deep Learning–SVM Classification Framework*

Kashan, Iran
Sep. 2015 – Sep. 2019

Research Interests

- ✓ Trustworthy AI
- ✓ LLM Evaluation & Reliability
- ✓ AI for Democratic Institutions
- ✓ Accountable AI Agents
- ✓ Human–AI Interaction
- ✓ Generative & Multimodal Models

Research Experience

✓ Researcher — Accountable AI Agents and Institutional AI Governance

May 2026 – Present

• In collaboration with *Dr. Mehrdad Tajbakhsh*

- Developing **accountability mechanisms** for autonomous agents: recording, explaining and defending their actions.
- Studying **Sovereign AI**: how institutional control over AI infrastructure shapes whether agents can be governed.
- Building an open-weight **LLM co-expert** for **intangible-asset valuation**, grounded in domain knowledge via **RAG**.
- Developing methods for **report reliability** and **data anonymisation** to protect user-disclosed information.

✓ Researcher at Qatar Computing Research Institute (QCRI)

March 2025 – March 2026

• Supervised by *Dr. Mohammad Amin Sadeghi, Dr. Anas A. Al-Nuaimi*

QCRI, Doha, Qatar (Hybrid)

- Core contributor to **Fanar**, an Arabic-centric multimodal generative AI platform; led culturally-aligned image generation.
- Built the platform's image-generation evaluation suite (FID, CLIP Score, Inception Score, PSNR, SSIM).
- Designed **15 custom LLM-as-a-judge metrics** spanning cultural accuracy and generation quality, informing model selection.
- Validated the judge scores against **Elo rankings from human pairwise comparison**, confirming reliability.
- Resulting manuscript: "Fanar Oryx: Culturally Aligned Image Generation for Arabic-Islamic Visual Culture" [5].

✓ Research Assistant at Computer Vision Lab

April 2024 – Present

• Supervised by *Dr. Mostafa Tavassolipour*

University of Tehran, Iran

- Building a hybrid **generative–algorithmic** assistant that proposes carpet patterns and components for designers to refine.
- Machine weaving requires **pixel-exact** patterns with a limited palette that standard generative models cannot produce.
- Pairing generative proposal with algorithmic constraint enforcement so outputs are directly usable in weaving production.

✓ Research Assistant at Computational Modeling and Machine Learning Lab

Jan. 2020 – March 2023

• Supervised by *Dr. Mehdi Modarressi, Dr. Mohammad Amin Sadeghi*

University of Tehran, Iran

- Developed "NU-Class Net", a custom deep neural architecture for enhancing low-bitrate compressed video quality.
- Achieved approximately $7\times$ video size reduction while preserving perceptual quality through a learned restoration network.
- Published in Engineering Applications of Artificial Intelligence (Elsevier, Q1) [1].

✓ Research Assistant at Intelligent Architectures for Computing Systems Research Lab

Nov. 2019 – Present

• Supervised by *Dr. Mehdi Modarressi*

University of Tehran, Iran

- Coordinating the lab's reading group on LLM hardware and efficiency.
- Optimizing GAN inference efficiency via pruning, quantization, and hardware-aware design.
- Designing energy-efficient LSTMs for wearable health devices, preserving accuracy under power and latency budgets [6].
- Co-authored a study on deep learning acceleration in 3D-stacked memories (IEEE CAL, 2023) [2].

✓ Artificial Intelligence Researcher

2018 – 2019

• Supervised by *Dr. Javad Salimi Sartakhti*

University of Kashan, Iran

- Built a hybrid DNN–SVM classifier with 5% accuracy gains over baselines; basis of B.Sc. thesis and paper [3].

✓ Augmented Reality Researcher

2017 – 2019

• Supervised by *Dr. Javad Salimi Sartakhti*

University of Kashan, Iran

- Developed a Unity-based AR application for surface detection and 3D object placement, validated across diverse object categories.

Publications

Peer-Reviewed

[1]

P. Z. Moghaddam, M. Modarressi, M. A. Sadeghi, “NU-Class Net: A Novel Deep Learning-based Approach for Video Quality Enhancement,” *Engineering Applications of Artificial Intelligence* (Elsevier, Q1), vol. 144, 2025. [doi:10.1016/j.engappai.2025.110118](#)

2025

[2]

S. H. S. A. Rezaei, **P. Z. Moghaddam**, M. Modarressi, “Smart Memory: Deep Learning Acceleration in 3D-Stacked Memories,” *IEEE Computer Architecture Letters*, 2023. [doi:10.1109/LCA.2023.3287976](#)

2023

[3]

P. Z. Moghaddam, J. S. Sartakhti, “Deep Learning Based on Support Vector Machines (SVM),” *5th National Conference on Distributed Computing and Big Data Processing*, 2019 (in Persian). [\[link\]](#)

2019

[4]

P. Z. Moghaddam, F. Lotfali, J. S. Sartakhti, “Improving Robot Behavior in a Real-Time Game Using the Grey Wolf Optimizer,” *2nd International Computer Games Conference: Opportunities and Challenges*, 2019 (in Persian). [\[code\]](#)

2019

Preprints & Manuscripts

[5]

P. Z. Moghaddam, K. Isufaj, M. A. Sadeghi, R. Abolhasani, A. A. Al-Nuaimi, “Fonar Oryx: Culturally Aligned Image Generation for Arabic-Islamic Visual Culture,” Qatar Computing Research Institute (QCRI); (Submission in preparation).

2026

[6]

F. Shayesteroodi, **P. Z. Moghaddam**, M. Mohammadinasab, M. Modarressi, M. E. Salehi-nasab, M. Daneshtalab, “Reducing the Complexity of Deep Learning Models for EEG Analysis on Wearable Devices,” 2026. [arXiv:2606.12742](#)

2026

[7]

P. Z. Moghaddam, M. Charbaghi, J. S. Sartakhti, “Faster Wav2Vec: Improving the Inference Speed of Speech Representation Models,” manuscript in preparation. Draft available upon request.

2026

Honors & Awards

- Awarded the **HATEF** Technology Development Grant, University of Tehran, selected from over 300 proposals.

2026
- Admitted twice to the Ph.D. program in Computer Science, University of California, Irvine (Fall 2024 & Fall 2025); unable to enroll due to U.S. visa processing delays.

2024 – 2025
- Awarded direct Ph.D. admission to the University of Tehran as an Exceptional Talent; declined to pursue a Ph.D. abroad.

2023 – 2024
- **Graduated 1st** (highest overall GPA) among M.Sc. Computer Engineering (CA) students, University of Tehran.

2022
- Awarded ‘**direct**’ (entrance-exam-exempt) **M.Sc. admission** to the University of Tehran as an **Exceptional Talent**.

2019
- Awarded ‘**direct**’ **M.Sc. Admission offer** from the [University of Kashan](#) and [Shahid Beheshti University](#) as an Exceptional Talent.

2019
- **Ranked 2nd** by GPA among Computer Software Engineering Students, University of Kashan.

2019
- **Recognized as a Distinguished Student** by The University Committee for the Exceptional Talents, University of Kashan

2018

Work Experience

- **Molavi Carpet: Software Engineer & Board Member**

2015 – Present

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Integrated multimodal generative AI into the production pipeline for product imagery and marketing videos.

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Leading the corporate web platform team and administering the company’s Linux servers and network infrastructure.
- **ModAI: Computer Vision Developer**

Jul. 2024 – Feb. 2025

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Developed object detection and instance segmentation models (YOLOv8, SAM-based) for fashion-product catalogs.

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Designed a custom image compression pipeline that reduced payload size by 75% and improved application response time.
- **TechnoFarsh (carpet-industry marketplace): Founder & Developer**

2017 – 2020

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Founded a B2B/B2C/C2C marketplace; incubated at the University of Kashan Innovation & Commercialization Center.

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Built a Unity-based AR application for virtual eyewear try-on and in-room carpet visualization.

Technical Skills

- **LLMs & Agents:** open-weight model deployment (Qwen3.6, FP8), LLM APIs (OpenRouter), RAG (LangChain), LLM-as-a-judge design, prompting strategies
- **ML & Systems:** model compression (pruning, quantization), distributed training (Slurm), generative metrics (FID, CLIP Score, IS, PSNR, SSIM), FFmpeg pipelines
- **MLOps & Infrastructure:** MLOps/LLMOps, Docker, Google Cloud Platform, Linux server administration, Git
- **Web & Deployment:** Django, FastAPI, Next.js, React, Node.js
- **Programming:** Python (PyTorch, NumPy, Pandas), C/C++, SQL, MATLAB | *also:* Julia, R, VHDL, Verilog
- **Languages:** English (IELTS Academic 7.5), Persian (Native)

Selected Teaching Experience

University of Tehran

Trustworthy AI (Dr. Mostafa Tavassolipour)	Head Teaching Assistant	2023 – Present
• Teach the course’s AI Ethics section; responsible for designing and grading all homework assignments and examinations.		
Generative Models (Dr. Mohammad Amin Sadeghi, Dr. Tavassolipour)	Teaching Assistant	2022 – 2023
Data Analysis (Dr. Mohammad Amin Sadeghi, Dr. Abolghasemi)	Teaching Assistant	2021, 2022

Neural Networks & Deep Learning (Dr. Ahmad Kalhor)	Teaching Assistant	2021
University of Kashan		
Machine Learning (Dr. Javad Salimi Sartakhti)	Head Teaching Assistant	2022
Discrete Mathematics (Dr. Ebrahimpour)	Head Teaching Assistant	2021
Artificial Intelligence (Dr. Ebrahimpour)	Head Teaching Assistant	2020

Selected Academic Projects

- **Denoising Diffusion Probabilistic Models (DDPM)**  2024
 - Implemented a DDPM from scratch for image denoising and generation, studying the forward/reverse diffusion process and the role of variance scheduling in sample quality.
- **Semantic Segmentation with U-Net**  2023
 - Developed a U-Net model for pixel-level segmentation of road, vehicle, and pedestrian classes on a self-driving car dataset.
- **Car Detection with YOLO**  2022
 - Built a YOLO-based detection pipeline for street imagery, producing bounding-box predictions on a real-world vehicle dataset.
- **LSTM-Based Music Generator**  2022
 - Trained a multi-layer LSTM for multi-genre music generation, exploring Delta-RNN and quantization for inference efficiency.
- **Bio-Inspired Computing : Optimization Algorithms**  2021
 - Implemented Genetic Algorithm, Particle Swarm Optimization, K-Means, and Ant Colony Optimization for classical problems including ZOE and Vehicle Routing.
- **Parkinson's Disease Classifier**  2021
 - Designed and benchmarked 15 classifiers (KNN, DNN, SVM, ensembles) on a 700+ feature clinical dataset, achieving 98.3% accuracy.
- **Generative Adversarial Networks (GAN) Suite**  2021
 - Constructed and compared DCGAN, C-GAN, SR-GAN, and WGAN architectures, analyzing convergence and output fidelity.

Selected Courses & Certificates

- **Creative Thinking Tools for Success and Leadership**  [Certificate], Imperial College London 2026
- **Google Prompting Essentials Specialization**  [Certificate], Google 2026
- **Machine Learning in Production**  [Certificate], DeepLearning.AI 2025
- **Machine Learning Specialization**  [Certificate], Stanford University 2024
- **2023 Oxford Machine Learning Summer School — Health**  [Certificate], OxML 2023
- **2023 Oxford Machine Learning Summer School — Finance & NLP**  [Certificate], OxML 2023
- **Deep Learning Specialization**  [Certificate], DeepLearning.AI 2023
- **Professional Neuroscience & Systems**, IPM School of Cognitive Science 2021 – 2022
- **Generative Adversarial Networks (GANs) Specialization**  [Certificate], DeepLearning.AI 2022
- **Deep Neural Networks with PyTorch**  [Certificate], IBM 2021

Workshops, Schools & Seminars

- **2024 Oxford MLx: Generative AI — Theory, Agents, Products**  [Certificate], AI for Global Goals 2024
- **8th Winter Seminar Series (WSS)**  [Certificate], Sharif University of Technology 2023
- **ReACT 2023**  [Certificate], Sharif University of Technology 2023
- **Scale TransformX**, Scale AI 2021 & 2022
- **How to be a Teaching Assistant**  [Certificate], University of Tehran 2022
- **Sixth IPM Advanced School on Computing & Artificial Intelligence**  [Certificate], IPM 2022
- **Fifth IPM Advanced School on Computing & Artificial Intelligence**  [Certificate], IPM 2021

References

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| <ul style="list-style-type: none"> • Dr. Mehdi Modarressi <ul style="list-style-type: none"> • ASSOCIATE PROFESSOR- UNIVERSITY OF TEHRAN ✉ modarressi@ut.ac.ir • Dr. Mohammad Amin Sadeghi <ul style="list-style-type: none"> • SENIOR SCIENTIST - QATAR COMPUTING RESEARCH INSTITUTE (QCRI) ✉ msadeghi@hbku.edu.qa | <ul style="list-style-type: none"> • Dr. Javad Salimi Sartakhti <ul style="list-style-type: none"> • ASSOCIATE PROFESSOR - UNIVERSITY OF KASHAN ✉ salimi@kashanu.ac.ir • Dr. Mostafa Tavassolipour <ul style="list-style-type: none"> • ASSISTANT PROFESSOR - UNIVERSITY OF TEHRAN ✉ tavassolipour@ut.ac.ir |
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